Power equations

1

a $x = 3$ **b** $x = 8$ **c** $x = \frac{8}{5}$ **d** $x = 6$

e $x = 2$ **f** $x = 6$ **g** $x = 5$ **h** $x = 2$

Exponential equations

2

a $x = 8$ **b** $x = \frac{2}{9}$ **c** $x = 3$ **d** $x = 0$

e $x = -2$ **f** $y = -3$ **g** $x = -3$ **h** $x = \frac{1}{3}$

i $n = \frac{1}{3}$ **j** $x = -2$

3

a

i $2^{3x} = 2^2$

ii $x = \frac{2}{3}$

b

i $2^{4x} = 2^{-1}$

ii $x = -\frac{1}{4}$

c

i $2^{-10} = 2^{2x}$

ii $x = -5$

d

i $2^{\frac{x}{2}} = 2$

ii $x = 2$

4

a $y = 4$ **b** $k = -2$ **c** $y = \frac{3}{2}$ **d** $x = 2$

e $x = 2$ **f** $x = \frac{7}{3}$ **g** $x = -7$ **h** $x = \frac{21}{19}$

i $y = \frac{11}{6}$ **j** $x = -\frac{41}{6}$ **k** $x = 5$ **l** $x = 1$

m $x = 1 + \frac{2k}{3}$ **n** $x = 5$ **o** $x = \frac{5}{2}$ **p** $x = 4, -1$

q $x = 3$ **r** $x = \frac{4}{1+n}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Exponential equations and logarithms

5

a

i $x = \dfrac{\log 5}{\log 13}$ ii 0.627

b

i $x = \dfrac{\log \frac{1}{11}}{\log 5}$ ii -1.490

c

i $x = \dfrac{\log 2}{\log 3}$ ii 0.631

d

i $x = \dfrac{\log 6.4}{\log 4}$ ii 1.339

e

i $x = \dfrac{\log 5}{\log 0.4}$ ii -1.756

f

i $x = \dfrac{\log 3125}{\log 5}$ ii 5

g

i $x = \dfrac{\log 6}{\log \frac{1}{2}}$ ii -2.585

h

i $x = \dfrac{\log 19\,207}{\log 27}$

ii 2.993

6

a

i Step 2, when factoring the a out of the index it should multiply the logarithm and not add to it.

ii $a = \dfrac{\log 40}{\log 9}$

iii 1.679

b

i Step 3. The 2 and 89 are in the incorrect positions in the logarithm.

ii $a = \dfrac{\log 89}{\log 2}$

iii 6.476

7 $x = -2$

Logarithmic equations

8

a $y = 16\,807$ **b** $y = 4$ **c** $y = 2$ **d** $y = 8$

e $y = 36$ **f** $y = 4$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

9

a 4 **b** 2 **c** $x = 81$ **d** $x = 100$

e $x = \dfrac{1}{1024}$ **f** $x = 2$ **g** $x = 8$ **h** $x = 5$

i $x = \dfrac{6556}{7}$ **j** $x = 15$ **k** $x = 6$ **l** $x = 32$

m $x = 137$ **n** $x = 4$ **o** $x = 2$ **p** $x = -10$

q $x = \dfrac{7}{5}$

10

a $x = 11$ **b** $x = 1,\ 6$ **c** $x = 16,\ 32$ **d** $x = 5$

e $x = 8$ **f** $x = \dfrac{343}{6}$ **g** $x = \dfrac{5}{16}$ **h** $x = 2 + \sqrt{11}$

i $x = 2$ **j** $x = \dfrac{81}{5}$ **k** $x = \dfrac{125}{4}$ **l** $x = 2$

m $x = -\dfrac{15}{7}$ **n** $x = 3$ **o** $x = 5$

11

a $y = -3$ **b** $y = \dfrac{1}{3}$

12

a $x = 2.15$ **b** $x = 0.13$

13

a $x = 9$ **b** $x = 0.05,\ 100$ **c** $x = 0.12$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647